Project Phase 3: Building a Resilient Database

Due Date: End of Week 5 **Points:** 100

Scenario

The Riverside Cafe's online ordering system is a huge success, but the single server running both the web application and the database is struggling. The order history has become a valuable asset, but the database is difficult to manage (patching, backups) and poses a performance bottleneck.

Your task is to propose and implement a new architecture that addresses these critical business and operational challenges by decoupling the database.

Project Goals

1. Propose a new database solution that offloads the heavy lifting of database administration.
2. Separate the database from the web application server to improve security and allow each layer to scale independently.
3. Execute a seamless migration of all existing order data from the current database to the new solution with a clear plan.

Technical Requirements

- The new database solution must be a managed service (e.g., Amazon RDS, Azure SQL Database).
- The database must be placed in a **private subnet** to prevent direct access from the internet.
- You must migrate all existing data. When the migration is complete, the application should function exactly as before.
- The necessary information regarding the previous database setup (e.g., schema, data dump utilities) has been provided to you by The Riverside Cafe's development team.

Submission Instructions

You will submit a single PDF document to Moodle. This document must contain all parts of the deliverable outlined below.

Deliverable Breakdown

- **Problem Analysis:** In your own words, summarize the key business and technical challenges The Riverside Cafe is facing with its current database setup.
- **Proposed Solution and Architecture:** Identify the type of managed relational database service that solves these problems. Justify your choice, explaining how it addresses concerns about maintenance, backups, and specialized skills. Provide a **"Before" Architecture Diagram** of your architecture from Phase 2 and a new **"After" Architecture Diagram** that reflects your redesigned

architecture. This diagram must show the web server, the new managed database instance in a private subnet, and the security group configurations allowing the connection.

- **Database Configuration Rationale:** What instance size did you choose for your database and why? Should you enable a high-availability feature (like Multi-AZ)? Justify your decision based on the trade-off between cost and reliability for The Riverside Cafe.
- **The Migration Plan:** Describe the step-by-step process you followed to migrate the data (mentioning tools like `mysqldump`). Explain how you reconfigured the web application to connect to the new database (e.g., updating the secret in Secrets Manager).
- **Verification:** How did you verify the migration was successful? Provide a screenshot of the "Order History" page *after* the migration, showing a new test order alongside all the previously migrated orders.

Grading Rubric

Criteria	**Description**	**Points**
Problem Analysis	Clear analysis of the existing problems and a well-justified proposal for a managed database solution.	10
Architecture Design	"Before" and "After" diagrams are clear and accurate. Rationale for instance size and high availability is well-reasoned.	30
Migration Plan	Migration process is clearly documented.	30
Verification	Reconfiguration and verification steps are explained, with screenshot proof of success.	20
Professionalism & Clarity	The submitted document is well-organized and clear. It also clearly discusses the team's working communication and individual contributions for the phase.	10
Total		**100**